

Can You *Prove* Your Agent Skill Works?

From vibe-checks to *evidence* — how to *actually* evaluate the agent skills you ship.

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EXHIBIT A — DOES IT HELP?

- ✓ valid output
- ✓ covers the cases
- ✓ well authored
- ✗ beats no-skill?

looks good.

But does it work?

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WORKING ON how teams turn real workflows into agent skills



IIT Bombay — Mech. Engg **9.90/10**, Institute Rank **2** · Minor in CS



Author of **Beyond Code** — a National Bestseller



Writes **Low-Pass Filter** — a signal-over-hype AI newsletter



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M365 CORE · INDIA DEVELOPMENT CENTER

■ THE GOLD RUSH

Everyone is making skills. Almost no one is checking them.

We automated skill creation. We never automated skill judgment.

~4k

skills on one public marketplace

SNYK • TOXICSKILLS

36.8%

were flawed on inspection

SNYK • TOXICSKILLS

13.4%

carried critical issues

SNYK • TOXICSKILLS

0

output-quality tests to publish

THE QUALITY GATE



The bar to publish, today: *"a SKILL.md and a GitHub account that's one week old. No code signing. No security review."*

*Anyone can make a skill. **Few can prove one earns its place.***

■ THREE QUESTIONS

“Is this skill good?” is really three questions.

And the industry answers them very unevenly.



Is it well-formed?

STRUCTURAL • STATIC

- Does the folder parse? Frontmatter valid?
- Links resolve, no secret leaking?
- Schema / manifest lint.



Is it good & safe?

INTRINSIC • STATIC

- Readable, complete, well-authored?
- Free of dangerous code patterns?
- Static scorers + security scanners.



Does it produce good output?

OUTPUT • DYNAMIC • RUNS IT

- Run it on real prompts — correct? complete?
- Better than the agent WITHOUT it?
- The one that matters. The one nobody does.

*Questions 1 & 2 never run the skill. **Question 3 does — and it's the verdict.***

Validation $\neq$ Evaluation

WHAT (ALMOST) EVERYONE CHECKS

- ✓ It parses & validates structural
- ✓ It reads well & looks safe intrinsic
- ✓ It follows the authoring style compliance

Cheap · automated · static. **Table stakes.**

WHAT ALMOST NOBODY CHECKS



Does it *actually* make the agent better at the task?

This is the **value proposition**. It needs you to run the skill on real tasks and grade what comes out — and it's the part people skip.

RUN IT · GRADE IT · COMPARE

Your skill passed lint. Did it pass the point?

Passing the schema tells you it's well-formed — nothing about whether it helps.

The real test: does it beat the model alone?

Two ideas do all the work — grade the outcome, and measure the delta.



Grade the outcome, not the path

An agent reaches a good answer in ways you didn't predict. Grade WHAT it produced — not how. Penalize the novel route and your evals break.



Measure the with / without delta

Run every case twice — with the skill, and without. It only earns its place if WITH scores meaningfully higher. The delta is what it buys vs. costs.

EXHIBIT — BENCHMARK DELTA



Δ +0.50 pass · +13s · +1700 tok

- ✓ +50 pts for +13s? ship it.
- ✗ 2× tokens for +2 pts? don't.
- passes with AND without? it's doing nothing.

Sometimes the skill makes it worse.

SkillsBench 2026 · 84 tasks · 11 domains · deterministic verifiers.

+16pp

avg lift from curated skills

BUT UNEVEN

16/84

tasks where skills went NEGATIVE

WORST -39pp

-1.3pp

from LLM-authored skills

≈ ADD NOTHING

2-3

focused skills beat "comprehensive"

MORE CAN HURT

What this proves

- "Does this skill help?" is falsifiable — and ~1 in 5 times the honest answer is no.
- **Eyeballing misses it — only the with/without delta catches a net-negative skill.**
- Models can't reliably author the skills they benefit from consuming.

How you actually evaluate output quality.

Not a research project — a tight, repeatable loop.

ANATOMY OF A TEST CASE



Prompt

A realistic user message — real paths, real context.



Expected output

Plain-English description of what success looks like.



Atomic assertions

One fact each, objectively checkable — added AFTER the first run.

GRADE: CHEAPEST METHOD THAT WORKS

- 1 **Deterministic verifiers** — valid JSON, row count, file exists
- 2 **Script checks** — mechanical, reusable, free
- 3 **LLM-as-judge** — boolean + 1-line evidence, schema-locked
- 4 **Human review** — for what assertions can't catch

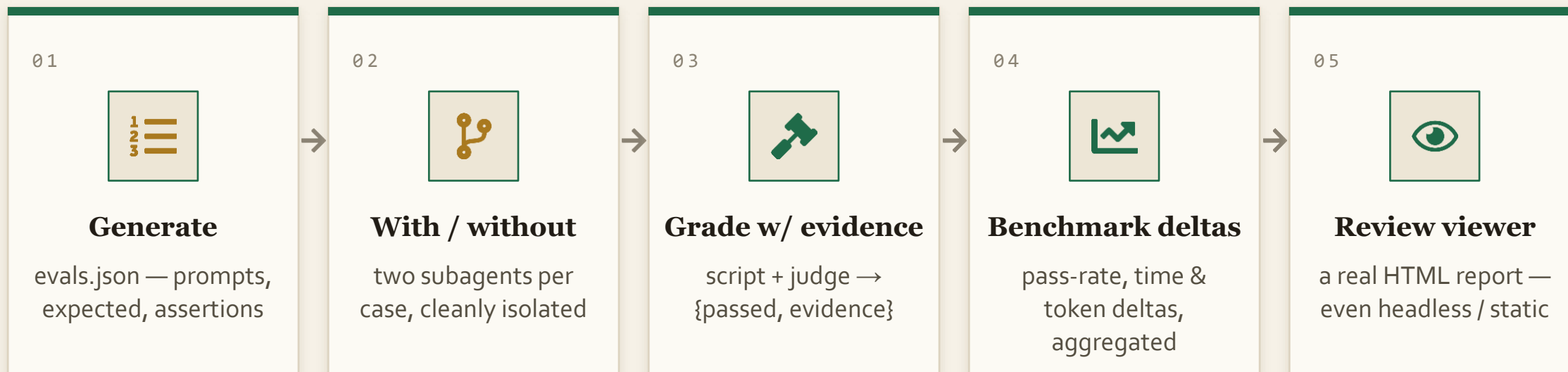


Run each case 3–5×

Same prompt gives different output run-to-run — don't trust one pass. Look at the distribution; high variance means a flaky eval or ambiguous instructions.

You don't build this harness. It already exists.

skill-creator — the same skill you author with — also runs the whole eval loop — BUT, on Claude Code.
So, I created *evaluate-skill-with-copilot-sdk*



The harness the industry “doesn't have” at publish time is built — and shippable — at authoring time.
Open source. Adopt the shape.

Let's put a real skill on trial.

A multi-step skill that encodes a real workflow — run through skill-creator, end to end.



The query set

5–6 realistic prompts + one edge case
— mined, not invented.



With vs. without, live

Two runs, same cases. Watch the
agent solve it both ways.



A finished run, ready

A full prior iteration's benchmark +
viewer — because real evals take time.

Where this is heading: automatic everything.

The 2026 research throughline — evaluation is the engine.



Automatic metrics

NO GROUND TRUTH NEEDED

- Score a skill with NO labels or test suites
- SkillAxe's 4 unsupervised dims: impact, trigger, compliance, coverage
- Emits a natural-language "improvement brief"



Automatic evaluation

NO STATIC BENCHMARK TO GAME

- Auto-build fresh tasks from real, evolving artifacts
- OpenSkillEval: 600+ instances across 30 community skills
- Maps a cost–performance frontier for skill selection



Automatic evolution

EVAL IS THE ENGINE

- Skills that rewrite themselves from eval signal
- +28% over un-refined skills; closes much of the gap to human-authored
- Every "evolution" method is an eval loop inside

Three things to take back



01

Stop vibe-testing

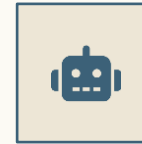
Treat skills as infrastructure, not prose. “It looked fine” is not a test — you can't build reliable systems on unverified capabilities.



02

Measure the delta

A skill that doesn't beat its baseline doesn't ship. ~1 in 5 are net-negative — only with/without catches it.



03

Let the harness do it

`evaluate-skill-with-copilot-sdk` or `skill-creator` runs the loop today; automatic metrics & evolution are coming. Adopt the shape now.

The verdict: Anyone can make a skill. From today, you can prove yours works.

That's the case. Thank you.

For more **AI** & **Data Science** — let's connect.



LinkedIn

`in/tezansahu`



Low-Pass Filter

`the signal-over-hype AI newsletter`